



DSYLEXIA FRIENDLY READER

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PROJECT OVERVIEW

The Dyslexia-Friendly Reader is a web-based application designed to help people with dyslexia read text more comfortably and efficiently. Many users with dyslexia face difficulties such as crowded text, complex words, poor spacing, and visual stress while reading digital content.

This project improves readability by:

- Breaking large text into smaller readable sections
- Increasing spacing between lines
- Simplifying difficult words
- Allowing users to customize reading settings
- Maintaining reading history for future access

The application provides a clean and accessible interface that makes reading easier and reduces cognitive load for users.

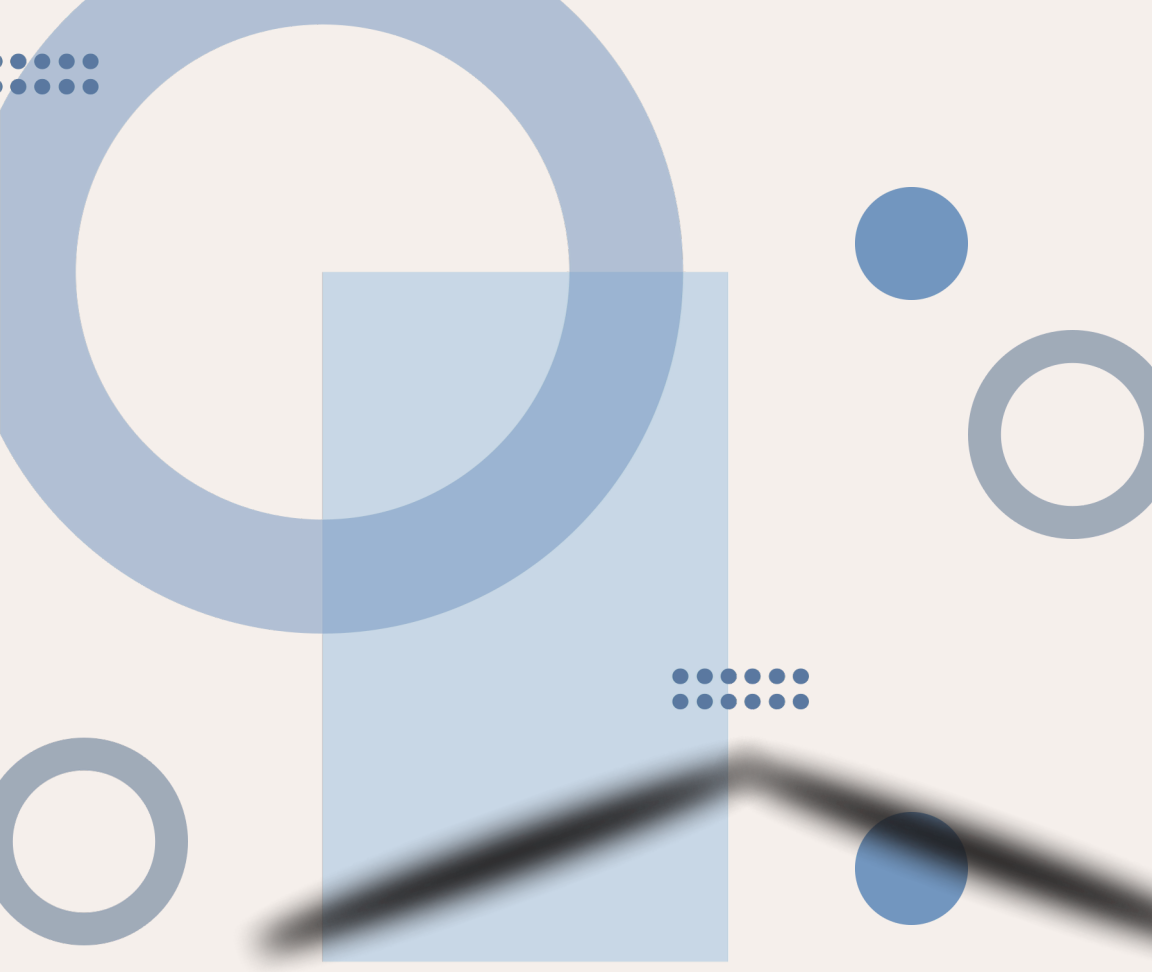


PROBLEM STATEMENT

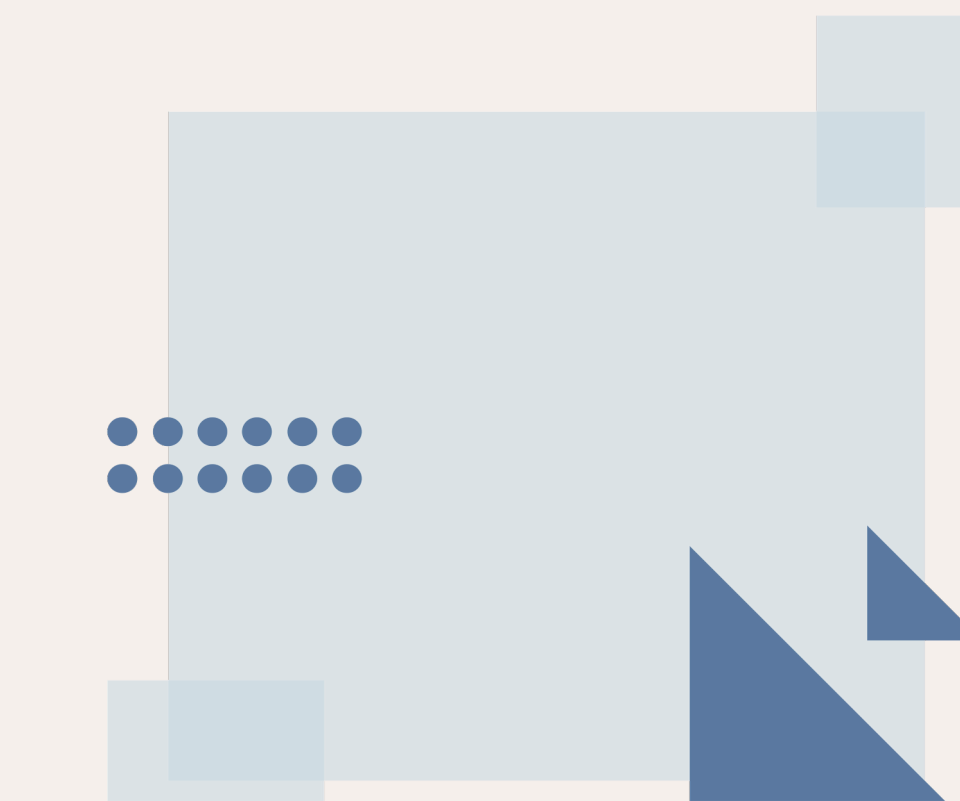
People with dyslexia often struggle to read standard digital text because of:

- Poor text spacing
- Complex vocabulary
- Crowded paragraphs
- Lack of personalization in reading interfaces

Most websites and reading applications are not optimized for dyslexic readers, making it difficult for them to understand and process information efficiently. The project aims to solve this problem by creating a customizable and accessible reading platform specially designed for dyslexic users.



OBJECTIVES



The main objectives of the project are:

To create a user-friendly reading platform for dyslexic users.

To improve readability through spacing and formatting adjustments.

To simplify difficult words into easier alternatives.

To allow users to customize their reading experience.

To provide login and history functionality for personalized usage.

To reduce reading stress and improve comprehension.

FEATURES

🔑 User Authentication

Signup and Login system

Session handling using localStorage

Logout functionality

📖 Dyslexia-Friendly Reading

Converts large text into readable chunks

Highlights separated sentences

Adjustable line spacing

🧠 Text Simplification

Replaces difficult words with simpler alternatives

Improves readability and comprehension

📊 Difficulty Checker

Analyzes text difficulty level

Displays:

Easy

Medium

Hard

⚙️ Custom Settings

Users can customize:

Font size

Letter spacing

Line spacing

Background color

📜 Reading History

Saves previously entered text

User-specific history storage

Delete individual history items

Clear all history

👤 Personalized Experience

Displays logged-in username

Separate history for each user

TECHNOLOGIES USED

Frontend

HTML5

CSS3

JavaScript

Storage

Browser LocalStorage

Development Environment

VS Code

XAMPP (for localhost testing)

ARCHITECTURE/WORLFLOW

Step 1: User Authentication

User creates an account using Signup page

Credentials are stored in localStorage

User logs in through Login page



Step 2: Session Management

Application checks login status

Logged-in user session is maintained



Step 3: Text Input

User enters or pastes text into the reader



Step 4: Text Processing

The system:

Splits text into smaller readable sections

Simplifies difficult words

Applies dyslexia-friendly formatting



Step 5: Difficulty Analysis

Word count is analyzed

Difficulty level is displayed



Step 6: Save History

User text is saved in personalized history



Step 7: Settings Customization

Users can adjust:

Font size

Line spacing

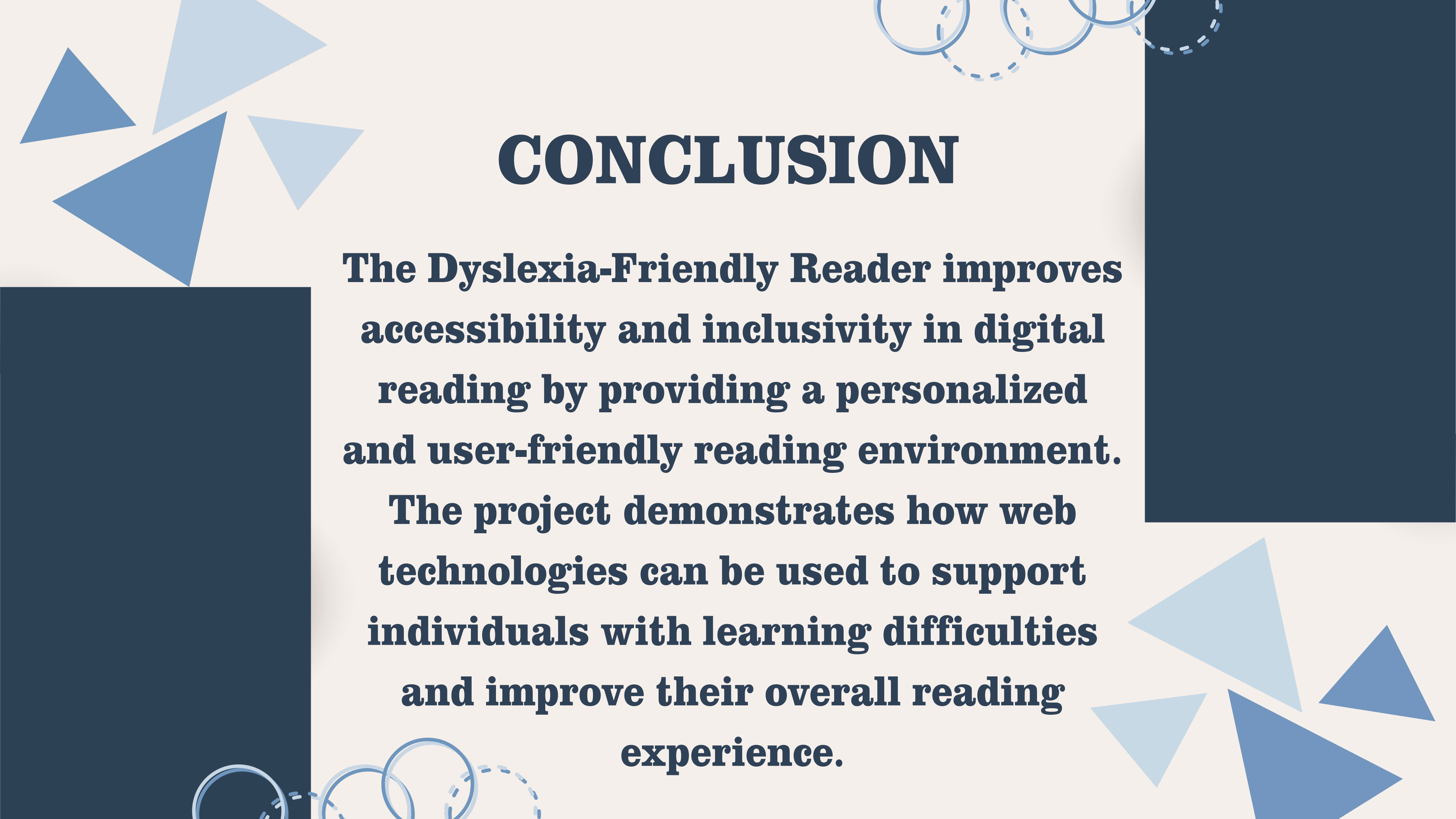
Letter spacing

Background color



Step 8: Output Display

Processed text is displayed in an easier-to-read format.



CONCLUSION

The Dyslexia-Friendly Reader improves accessibility and inclusivity in digital reading by providing a personalized and user-friendly reading environment.

The project demonstrates how web technologies can be used to support individuals with learning difficulties and improve their overall reading experience.



THANK
YOU!